

Research Proposal Review

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Topic: Cybersecurity threats in IoT for the Power and Utilities sector (smart grids and SCADA)

1) Which of the methods described in this week's reading would you think would suit your purpose?

- Choose a mixed methods approach to blend in depth insights with practical evidence.
- Add an exploratory qualitative part to uncover threat routes, work limits, and choice reasons in OT-IT setups.
- Link it to a descriptive or confirmatory quantitative part to check patterns and measure how certain controls affect risk signs.
- Join several case studies with small tests in a mock substation or AMI setup to meet safety rules.
- Cross check between parts so findings from talks and papers shape survey questions and test measures.

2) Which data collection methods would you consider using?

- Hold semi structured talks with SCADA experts, OT security heads, and grid managers to note real work and limits.
- Run a quick survey for workers to see how often certain threats and controls show up in power firms.
- Review papers on zone maps, entry rules, upkeep steps, and event guides to trace attack routes and safeguards.
- Record mock OT data flows (IEC-104, DNP3, Modbus/TCP) to evaluate block lists and odd spotting without real systems.
- Map out studies from experts and key guides (like NIST SP 800-82; IEC 62351; ENISA IoT safety) to build a solid fact base.

3) Which required skills will you need to have or develop for the chosen project?

- Study setup: mixed methods plan, case guides, and clear reports.
- Qualitative tools: talk plans, note taking, theme sorts, and feedback checks.
- Quantitative tools: test setups, simple data stats, impact measures, and repeat methods.
- Field ability: OT/ICS basics, zones and links, life cycle limits, and protocol skills (Modbus/TCP, DNP3, IEC-104; safe setups in IEC 62351).
- Data rules and safety: consent forms, name removal, secure holds, and full split of test data from live ones.
- Sharing skills: short school writing, easy charts for workers, and quick overviews for non-experts.